No-tools trace (baseline)

1 turn · 0 tool calls · hindsight description

user: panoramic X-ray, tool-free prompt

"identify any abnormal teeth and diagnose"

thought (single turn)

"On systematic review of this panoramic radiograph, I begin by orienting to the patient's right and left sides as marked. The image shows a mixed dentition with multiple developing permanent teeth ...

Focusing on the lower-left quadrant (Q3), I examine the posterior region ... I can see a radiolucent area involving the occlusal/cervical portion of the tooth, with loss of normal enamel contour and a darker zone consistent with a carious lesion ..."

One-shot description from the full image: no crop, no second look, nothing checkable against the pixels it actually saw.

final answer

Q3T4 · Caries · confidence 0.82

What this arm can teach

- output format + clinical vocabulary
- quadrant-survey habit

No evidence trail to verify or correct.

With-tools trace (main policy)

23 turns · 22 real tool calls · 3 corrective nudges

T0	**window_level**	survey the full field at high contrast
T1	**fdi_label**	orient to FDI notation first
T2	**locate_tooth(34)**	box [1668, 917, 84, 322] (GT-grounded detector)
T3	**zoom_crop**	"shows only bone and root area ... too low"
T4	**nudge_crop #1**	dy -0.3 · scale 0.6 -> [1685, 885, 50, 193]
T6	**nudge_crop #2**	dy -0.5 · scale 0.7 -> [1693, 818, 35, 135]
T8	**nudge_crop #3**	dy -0.6 · scale 0.8 -> [1696, 750, 28, 108]
T9-10	**zoom + denoise**	"Excellent! Now I can clearly see tooth 34"
T14+	**compare + enhance**	contralateral check, contrast pass, re-verify

final answer (turn 22) · Q3T4 · Caries · conf 0.82

matches ground truth - passed strict verification

22 real tool calls · 3 corrective · every observation executed against the actual radiograph

What the exact 1:1 pairing isolates

same image · same prompt template · same ground truth - only the tool channel differs

with-tools traces carry 44,350 executed tool calls; the no-tools arm carries zero

5 cohorts × 2 arms · 1,847 + 1,847 · 3,694 verified traces

any SFT delta is attributable to demonstrated tool use, not to data volume or prompt wording

per-cohort counts match exactly (678/27/660/202/280 on each side) - parity by construction